Radical Pair Hamiltonian Parameter Comparison

Convention: all Hamiltonian coefficients are reported as frequencies, A/h, J/h, and |D|/h in MHz. The point-dipole prefactor is |D|/h = 52.04 MHz nm^3 / r^3. Some OOP-ESEEM papers report a different dipolar D convention, approximately D_OOP = -3 D / 2; this table uses |D|/h to avoid ambiguity.

System	Radical pair	r (A)	J/h (MHz)	|D|/h (MHz)	Dominant A/h (MHz)	Notes / provenance
Cryptochrome	FAD radical anion / TrpD radical cation	21.0 ErCry4; 21.2 +/- 0.2 pigeon Cry4	~0.03	5.6; pigeon 5.46 +/- 0.15	FAD N5 principal: -2.2, -2.0, 47.5. Trp Hbeta1: 42.0, 43.7, 49.2. Trp N1 max: 30.3.	Distance and J from OOP-ESEEM literature. Hyperfine values from flavin/Trp radical datasets. Strong A terms are comparable to or larger than D.
Photolyase	Measured photoreduction pair: FAD radical anion / TrpC radical cation. Repair pair: FADH radical / CPD radical anion.	FADH/CPD repair pair: ~4-11	Not directly measured; close-contact estimate ~10^1-10^3, likely >>0.05 FAD-TrpC. Use as inferred regime, not a fitted coefficient.	closest-contact upper-bound ~500-600	Measured FAD-TrpC pair: use FAD/Trp values above. Repair-pair CPD- proxy: one-electron reduced thymine/oligo(T) ESR best-fit A(H6)=(16.01, 35.66, 57.88) MHz, Aiso=36.5 MHz; weak Aiso estimates H3 4.0, H1' 2.0, N1 2.5, N3 0.7, CH3 3.7 MHz. Treat as a thymine-radical-anion surrogate, not a direct CPD tensor.	FAD-TrpC OOP-ESEEM values describe photoactivation, not the repair pair. For the FADH/CPD repair pair, structural FAD-to-lesion distances from 2XRZ/5ZCW put it in a close-contact regime: dipolar coupling is tens to hundreds of MHz and exchange should be much larger than the TrpC pair, but no direct fitted J was found.
Photosystem I / photosynthetic RC	P700 radical cation / A1 radical anion	25.4 +/- 0.03	Likely << |D|; often unresolved/small for long-lived separated pair	3.18 +/- 0.01	P700+ donor: dominant 1H and 15N couplings, max |A| ~6 MHz; Aiso up to ~3.5 MHz. A1- acceptor: nearby 15N couplings weaker, Aiso ~0.3-2.0 MHz; anisotropic terms up to ~0.6 MHz.	Regularized summary from measured/fitted ENDOR/ESEEM tensors. Source tables report axial/principal tensor forms; this cell lists comparable max |A| and Aiso-scale values.
Bacterial reaction center	P865 radical cation / QA radical anion	28.4	Likely small for separated pair	2.27	P865+ donor: dominant 1H couplings max |A| ~10 MHz in P866 state, up to ~14 MHz in P850-like conformation; 15N max |A| ~6 MHz, Aiso up to ~3.7 MHz. QA- model-B tensors: 17O O1 (-95,19,19), O4 (-80,13,14); 13C C4 (41,-9,-11), C1 (-20,-16,-11) MHz.	Regularized summary from measured/fitted ENDOR/ESEEM donor tensors. Uploaded papers strongly cover P865+; QA- values are DFT model-B total principal tensors for the Rb. sphaeroides QA site, benchmarked against ENDOR data.
Carotenoid-porphyrin-fullerene triad	Carotenoid radical cation / fullerene radical anion	Estimated radical-centroid distance ~35-45 A (carotenoid C+ spin-density centroid to C60-cage centroid); plausible geometry range ~30-45 A.	Neglected in published CPF spin-dynamics models; not fitted. At this separation, exchange is treated as small.	Point-dipole estimate ~0.6-1.9 for r = 45-30 A (about 1.0 at 37 A); published CPF models neglect dipolar coupling.	Carotenoid 1H isotropic A/h: largest +16.23, -10.81, -10.12, +9.22 x3, -8.88, +8.55 x3 MHz; 45 proton couplings, RMS 39.1 MHz. C60 treated hyperfine-free.	Lindoy/Lewis CPF model uses g = ge, ignores 13C, and uses kS = 1.8e7 s^-1, kT = 7.1e4 s^-1. Scalar exchange and anisotropic dipolar coupling are explicitly neglected; r and |D|/h here are structure-derived estimates, not measured/fitted coefficients.
AsLOV2 WT	Triplet FMN / C450 contact intermediate leading to cysteinyl-flavin adduct	Likely <10 A contact scale	Strong/contact; not cleanly measured as weakly coupled RP	>52 if r < 1 nm; ~416 at r = 0.5 nm	Flavin neutral examples: H5 -21.6; N5 11.0; N10 5.9.	Strong-coupling comparator. The row should be framed as contact flavin-cysteine photochemistry, not a measured separated RP Hamiltonian.

MagLOV	Likely FMN radical anion / protein donor radical cation; later FMNH radical chemistry	Donor not pinned	Not measured	Not measured	If Trp donor: Hbeta1 ~45. If flavin anion: N5 principal max 47.5.	Excellent MFE/ODMR sensor row, but microscopic J, D, and donor geometry are not yet measured in the same way.

Source Notes

- OOP-ESEEM distances/J discussion: Al Said, Weber, and Schleicher, Front. Mol. Biosci. 2022, doi: 10.3389/fmolb.2022.890826.
- RadicalPy hyperfine database and supplement: Evans et al., J. Chem. Theory Comput. 2024, doi: 10.1021/acs.jctc.4c00887.
- Chemical compass triad benchmark: Maeda et al., Nature 2008, doi: 10.1038/nature06834.
- MagLOV source: Abrahams et al., bioRxiv 2025, doi: 10.1101/2024.11.25.625143.
- CPF hyperfine/model parameters: Lewis et al., J. Chem. Phys. 2014, doi: 10.1063/1.4890659; Lewis et al., J. Chem. Phys. 2018, doi: 10.1063/1.5038558; Lindoy et al., J. Chem. Phys. 2020, doi: 10.1063/5.0006411.
- Photosystem I hyperfines: Kaess et al., Chemical Physics 1995, doi: 10.1016/0301-0104(95)00021-F; Kaess and Lubitz, Chem. Phys. Lett. 1996, doi: 10.1016/0009-2614(96)00129-7; ESEEM study of A1 phyllosemiquinone, Biochemistry 1997, doi: 10.1021/bi971360a.
- Bacterial RC hyperfines: Kass et al., J. Phys. Chem. 1995, doi: 10.1021/j100001a065; Rautter et al., Biochemistry 1997, doi: 10.1021/bi962859s; Lendzian et al., Photosynth. Res. 1998, doi: 10.1023/A:1006030221445.
- QA- ubisemiquinone hyperfine source: O'Malley, Chem. Phys. Lett. 1998, doi: 10.1016/S0009-2614(97)01456-5; reports B3LYP/EPR-II model-B QA-site 17O and 13C total principal hyperfine tensors in MHz.

- CPD repair-pair proxy hyperfines: Bernhard and Patrzalek, Radiation Research 1989, doi: 10.2307/3577345; ESR parameter set for one-electron reduced thymine T- in oligo/poly(T), with A(H6)=(16.01, 35.66, 57.88) MHz and weak Aiso estimates H3 4.0, H1' 2.0, N1 2.5, N3 0.7, CH3 3.7 MHz. Used here only as a CPD- surrogate.

- FADH/CPD repair-pair geometry: RCSB PDB 2XRZ and 5ZCW, M. mazei class II CPD photolyase complexes with intact CPD lesions. Distances estimated from the FAD N5/C4X midpoint to CPD O4T / lesion-base centroid give ~8.5-11.6 A; closest isoalloxazine-lesion contacts are ~4.4-4.8 A. The tabulated |D|/h values are point-dipole estimates from this range. Exchange for this repair pair is not experimentally fitted here; the listed J/h range is an order-of-magnitude close-contact inference.

- CPF distance estimate: no explicit measured C+/C60- radical-centroid distance was found in Maeda/Lewis/Lindoy spin-dynamics sources; estimate is based on the published C-P-F triad geometry/extended molecular structure, using the carotenoid radical-cation spin-density centroid and the fullerene cage centroid. Treat as order-of-magnitude only; |D|/h is computed from |D|/h = 52.04 MHz nm^3 / r^3.